

MIHIR APTE

Data Scientist | AI/ML Engineer | MSc Computer Science (Data Science) @ Trinity College Dublin

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Work Authorisation: Stamp 1G (permitted to work in Ireland)

SUMMARY

MSc Computer Science (Data Science) graduate (Trinity College Dublin) who has independently scoped, built, and shipped five end-to-end AI and machine learning projects, spanning NLP, generative AI, classical ML, and physics-informed deep learning, each delivered with a working demo, interactive dashboard, or deployed app so results are usable by someone else, not just a notebook. Comfortable formulating a problem from scratch, building a reproducible evaluation pipeline, and **being honest about where a metric is hiding something important**. Strong Python foundations across ML, NLP, and scientific computing frameworks (PyTorch, HuggingFace, BERTopic, scikit-learn, XGBoost) and backend engineering (FastAPI, Flask). Background also includes three internships spanning data engineering, data science, and full-stack software development.

EDUCATION

MSc Computer Science, Data Science, Trinity College Dublin

Sep 2025 to Aug 2026

Modules: Machine Learning, Data Analytics, Artificial Intelligence, Data Visualisation, Text Analytics, Scalable Computing, Information Retrieval, Security and Privacy

BE Information Technology, Savitribai Phule Pune University

May 2020 to Jun 2024

Marathwada Mitra Mandal's College of Engineering, Pune | CGPA: 9.02 / 10.0 | Modules included Engineering Mathematics, Numerical Methods, and core Physics

RESEARCH & PROJECTS

MSc Dissertation: Improving RAVE Video Editing Model

Trinity College Dublin, 2025 to 2026

- Improved temporal consistency of a published CVPR 2024 zero-shot video-editing model, as measured by **Warp Error and CLIP Score benchmarked across 21 test videos against 4 competing methods**, by formulating and implementing two new frame-grouping algorithms in PyTorch (CLIP ViT-B/32 similarity and K-means clustering) to replace the model's random frame-shuffling step
- Delivered a reproducible, quantitative evaluation pipeline for a generative computational model, as measured by **consistent benchmark results held across richer structural conditioning** (Multi-ControlNet: ZoeDepth, Canny edge, FreeU), by engineering DDIM caching and GPU-compatibility fixes across Google Colab and Kaggle cloud compute environments
- Isolated which structural-conditioning signal actually drove the improvement, as measured by **an ablation across ZoeDepth, Canny edge, and FreeU run independently and in combination**, by designing a controlled evaluation protocol that varied one conditioning signal at a time while holding the frame-grouping method fixed

Tools: PyTorch, HuggingFace Diffusers, CLIP ViT-B/32, ControlNet, Stable Diffusion 1.5, K-means, OpenCV, Python, Git

Physics-Informed Neural Network for Heat Conduction

Self-directed project, completed

- Trained a neural network to solve heat-conduction PDEs with no labeled training data, as measured by **matching a closed-form analytical solution to within 0.0015°C** (under 0.002% of the imposed 100°C boundary difference), by implementing a physics-informed loss (PDE residual plus boundary conditions) computed via second-order PyTorch autograd and a two-stage Adam-then-L-BFGS optimisation scheme
- Extended the solver from a 1D steady-state baseline to the 2D (space and time) transient heat equation, as measured by **under 1% max relative error reproducing a multi-mode decaying temperature field** across the full space-time domain, by designing a closed-form multi-eigenmode benchmark and adding an initial-condition loss term alongside the PDE and boundary terms
- Cross-validated the model against two independent ground truths instead of trusting a single derivation, as measured by **a custom finite-difference solver agreeing with the closed-form solution to about 1e-4**, by implementing and stability-analysing an explicit FTCS scheme from scratch
- Diagnosed the specific driver of model error rather than reporting an aggregate metric, as measured by **isolating a near-zero correlation (0.01) between local error and solution gradient magnitude**, by running a gradient-vs-error correlation analysis and a controlled collocation-point ablation (30 of 1000 points cut error 50x) that pinpointed proximity to the initial condition, not local sharpness, as the true cause

Tools: Python, PyTorch (autograd, L-BFGS), NumPy, Matplotlib, Jupyter, finite-difference methods, Git and GitHub

Comment Lab: AI Content Assistant for Creators

Phase 1 (YouTube) shipped, Phase 2 (Instagram) in progress

- Shipped a full-stack AI content assistant for creators, as measured by **zero-setup installation for non-technical external testers**, by packaging an end-to-end YouTube ingestion, topic-modelling (BERTopic, UMAP, HDBSCAN), and multilingual sentiment pipeline as a one-click Windows installer (PyInstaller, Inno Setup) that silently provisions a local LLM runtime on first run
- Made audience data actionable for non-technical creators, as measured by **a four-tab interactive dashboard (Topics, Trending Matches, Content Requests, Content Ideas) built for on-demand review**, by designing and shipping a FastAPI-backed data visualisation layer on top of the underlying NLP pipeline
- Reduced content-idea guesswork to an evidence-backed process, as measured by **every suggestion citing the real comment volume and sentiment behind it**, by building a two-stage request detector (regex heuristic plus local LLM confirmation) and cross-referencing discovered topics against live YouTube trending data
- Managed the project's data-privacy risk directly, as measured by **fully isolated, namespaced storage per creator and per tester with zero cross-contamination**, by designing a multi-tenant, privacy-by-design data architecture with frozen-app-aware path resolution
- Took ownership of project scope to ship on a realistic timeline, as measured by **a documented set of deliberate scope cuts (CSV-backed storage instead of a database, a single static dashboard instead of a full frontend framework)**, by prioritising a right-sized Phase 1 release over an over-engineered one
- Planned a second phase before writing any integration code, as measured by **an evaluated integration path reusing the existing topic-modelling and sentiment architecture**, by researching Meta's Instagram Graph API and OAuth access-tier tradeoffs for Phase 2

Tools: Python, JavaScript/HTML/CSS, FastAPI, BERTopic, UMAP, HDBSCAN, HuggingFace Transformers (XLM-RoBERTa), scikit-learn, Ollama (Qwen2.5), YouTube Data API v3, YouTube Analytics API, Google OAuth 2.0, PyInstaller, Inno Setup, Git

Cognitive Distortion Detector [Live App](#)

Deployed: HuggingFace Spaces

- Automated detection of **10 cognitive distortions** from free text, as measured by validation against a licensed psychologist's expert annotations, by gathering domain requirements directly from a subject-matter expert and fine-tuning DistilBERT and RoBERTa transformer classifiers on synthetic training data generated from scratch (no public dataset existed)
- Improved decision reliability in a sensitive domain, as measured by **eliminating high-confidence errors on neutral text** (RoBERTa scored 0.981 confidence on a false positive), by selecting a properly calibrated threshold and adding LIME explainability so every prediction is clearly communicable to non-technical reviewers

Tools: Python, PyTorch, HuggingFace Transformers (DistilBERT, RoBERTa), LIME, scikit-learn, Streamlit, Git

Customer Churn Prediction & Business Simulator [Live App](#)

Deployed: HuggingFace Spaces

- Reduced the expected financial cost of churn misclassification, as measured by **optimising the decision threshold from a default 0.5 to an evidence-based 0.11** against a real, risk-adjusted cost function (12 months' lost revenue vs. about €50 false-alarm cost), by building and automating an end-to-end XGBoost optimization pipeline
- Enabled non-technical stakeholders to independently interrogate model decisions, as measured by **real-time probability and explanation updates**, by building an interactive SHAP-powered what-if simulator in Streamlit for clear stakeholder communication
- Ensured robust model evaluation despite severe class imbalance, as measured by **AUC-ROC and precision-recall curves** rather than accuracy alone, by applying SMOTE resampling and automating threshold-sensitivity testing across the full probability range

Tools: Python, XGBoost, SHAP, SMOTE, scikit-learn, Pandas, Streamlit, HuggingFace Spaces, Git

EXPERIENCE

Data Engineer Intern, C4i4 Labs

Dec 2024 to May 2025

Pune, India · On-site

- Reduced manual effort in day-to-day data retrieval, as measured by **faster, self-service query resolution for non-technical users**, by engineering and automating an NLP-based conversational chatbot in Python for clear stakeholder communication
- Surfaced actionable customer-satisfaction insights for the business, as measured by **sentiment trends across the full feedback corpus**, by building and automating an NLP-based data pipeline on customer feedback data
- Streamlined candidate-job alignment for recruiters, as measured by **automated parsing of unstructured CVs against job requirements**, by developing an AI-powered resume-matching system using NLP
- Improved operational efficiency for automotive service operations, as measured by **automated, optimization-driven scheduling recommendations**, by designing an AI-driven scheduling and process-automation system that replaced a previously manual, ad-hoc operations workflow

Data Scientist Intern, AlgoAnalytics

Jul 2024 to Dec 2024

Pune, India · Remote

- Enabled real-time risk monitoring for a Solar Panel Digital Twin, as measured by **continuous anomaly detection on live sensor data**, by building and automating a predictive-maintenance model in Python and scikit-learn

- Reduced stakeholder review time on large PowerPoint presentations, as measured by **condensed, decision-ready summaries**, by developing an LLM-powered document-summarisation module using Hugging Face for clear stakeholder communication

Software Developer Intern, Invasystems Inc.

Feb 2023 to May 2023

Pune, India

- Streamlined candidate-pipeline management for recruiting teams, as measured by **automated tracking of every application stage**, by gathering the recruiting team's requirements and building a full-stack Application Tracking System using Python, Flask, and SQL that replaced a previously spreadsheet-based workflow

TECHNICAL SKILLS

Data Science, ML & NLP: PyTorch, TensorFlow, Keras, HuggingFace Transformers/Diffusers, BERTopic, UMAP, HDBSCAN, Sentence-Transformers, DistilBERT, RoBERTa, scikit-learn, XGBoost, SHAP, SMOTE, LangChain, Ollama, LIME, OpenCV, torchvision

Optimisation, Numerical Methods & Scientific Computing: physics-informed neural networks, PDE residual loss design, finite-difference methods, cost-function-driven decision optimisation, L-BFGS and Adam optimisation, clustering-based algorithm design (K-means), threshold and hyperparameter search, SHAP/LIME explainability, numerical evaluation pipelines

Data Visualisation & BI: Streamlit, custom FastAPI/JS dashboards, SHAP visual explainability, Matplotlib, Seaborn, Power BI, Tableau

Programming & Backend Engineering: Python, JavaScript, HTML/CSS, SQL, FastAPI, Flask, Git, REST APIs, OAuth 2.0

Tools & Deployment: PyInstaller, Inno Setup, Gradio, HuggingFace Spaces, Jupyter, Pandas, NumPy, SciPy, Google Colab and Kaggle (cloud GPU environments)

CERTIFICATIONS

- SQL for Data Analytics using PostgreSQL (Udemy)
- Google AI-ML Program (AICTE)
- Intel AI/ML Certificate
- Python for Data Science and Machine Learning (Udemy)